

Homework

EVOLVING DOCUMENT

Plaksha: Robot motion simulation and optimization

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1 Problems

1. **Matlab.** Master all parts of the Pratap Matlab book that we posted on Moodle
2. **Numerical derivative.** Use the definition of derivative to calculate the derivative of $\sin(x)$ at $x = 1$.
 - (a) Compare that to the exact value (exact value is $\cos(1)$). [Practically speaking, ‘exact’ means the full computer precision before the compounding of roundoff errors. That is, it is the roundoff error of numbers and of elementary functions.]
 - (b) Plot error vs h (log log plot) and note that there is a best h because of the competition between Method error and roundoff error.
 - (c) Challenge: Using the Matlab resolution of Matlab’s smallest number “eps” (look it up), estimate the best h for this calculation and compare It with the minimum on your graph.
3. **Tutorial problem.** Complete the exercises from the Tutorial on Thursday morning 29 January.
4. **Euler solver.** Demonstrate that you can write code equivalent to the code shared with you on 30 January. That code separates the solution of an ODE into
 - i) a script (problem set up) and two matlab functions:
 - ii) The Right Hand Side (RHS) function; and
 - iii) The Euler solver (that can solve any set of first order ODEs). If you do that first, you can use it to make the error plots above. You should be able to write this using Matlab help and no more (no Google, no GPT, no copilot).
5. **Make up a problem.** Cook up, or find, some ODE, or set of first order ODEs, and solve them and explore something hopefully-interesting about the solution.

6. Higher order methods.

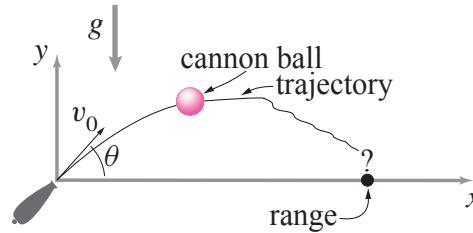
- (a) Calculate the derivative of $\sin(x)$ using central difference. Make a log log plot of error vs h . On one plot show Euler's method and the central difference method. You should notice that
- the best h is slightly bigger with central difference than Euler
 - for given h the error is much smaller with central difference
 - the slope of the curve for central difference is about 2, twice the slope of the curve for Euler (about 1). This indicates that the error for Euler is proportional to h , and for central difference the error is proportional to h^2 . Hence Central Difference is a so called 'second order' method, whereas with Euler, the slope is one indicating that error is proportional to h so it is called a 'first order' method
- (b) Write a new Matlab function that solves arbitrary ODEs using 'the midpoint method'. This is only a slight modification of the Euler solver. Test it with the differential equation $\dot{x} = x$ with initial condition $x(0) = 1$, evaluated at $t = 1$. On one log log plot show the error as a function of h for both the Euler method and the midpoint method.

7. **Spring and mass (2D).** One end of a negligible-mass spring (k, L_0) is pinned to the origin, the other to a mass (m). There is gravity (g). Initial Conditions (ICs): The initial position is $\vec{r}_0 = x_0\hat{i} + y_0\hat{j}$, and the initial velocity is $\vec{v}_0 = v_{x0}\hat{i} + v_{y0}\hat{j}$. Motion starts at $t = 0$ and ends at t_{end} .

- (a) Find the Equations of Motion (EoM);
- (b) Assume all parameters and IC's above are given.
- Plot the trajectory of the mass.
 - Animate the trajectory of the mass.
- (c) How many ways can you think of checking the numerical solution? Find as many as you can, and do the check. A list of things to check is started here:
- $k = 0$, all else arbitrary: The motion is parabolic flight (including falling straight down as a special case) [Why? The system is then just ballistics from freshman physics];
 - $x_0 = 0, v_{x0} = 0$, all else arbitrary: The motion stays on the y axis [Why? There is no force in the x direction if the mass is on the y axis. Because the initial velocity has no x component, the mass never leaves the y axis];
 - $g = 0, \vec{v}_0 = \vec{0}$, all else is arbitrary: The motion stays on a radial line. And, if the motion does not cross the origin, the motion is that of a harmonic oscillator (sinusoidal oscillations, check by plotting, say x vs t . [Why? (Advanced) Write the EoM and EoMs in polar coordinates
 $\Rightarrow m\ddot{r} = -k(r - L_0)$
 $\Rightarrow$ the harmonic oscillator equation, $m\ddot{r}^* = -kr^*$, where $r^* \equiv r - L_0$].
 - $L_0 = 0$, all else is arbitrary: ____? _____. [Why? ____? _____.] The question marks indicate the interesting features of the solution and the explanation for those interesting features.
 - etc.*

- vi. *etc.*
- vii. ...

8. **Canon ball.** A cannon ball m is launched at angle θ and speed v_0 . It is acted on by gravity g and a viscous drag with magnitude $|cv|$.
- (a) Find position vs time analytically (optional, for those who like theory)
 - (b) Find a numerical solution using $\theta = \pi/4$, $v_0 = 1 \text{ m/s}$, $g = 1 \text{ m/s}^2$, $m = 1 \text{ kg}$, $c = 1 \text{ kg/s}$.
 - (c) Compare the numeric and analytic solutions. At $t = 2$ how big is the error? How does the error depend on specified tolerances or step sizes?
 - (d) Use larger and larger values of v_0 and for each trajectory choose a time interval so the canon at least gets back to the ground. Plot the trajectories (using equal scale for the x and y axis. Plot all curves on one plot. As $v \rightarrow \infty$ what is the eventual shape? [Hint: the answer is simple and interesting.]
 - (e) For any given v_0 there is a best launch angle θ^* for maximizing the range. As $v_0 \rightarrow \infty$ to what angle does θ^* tend? Justify your answer as best you can with careful numerics, analytical work, or both.



9. **Statics vs Dynamics.** 2D. Equilibrium and dynamics of a point mass m at $\vec{r}_C$. Gravity g points down in the minus $\hat{j}$ direction. Points A and B are anchored at $\vec{r}_A$ and $\vec{r}_B$. Springs and parallel dashpots connect the mass to A and B. The spring constants are k_A and k_B . Their rest lengths are L_A and L_B . Parallel to the springs are dashpots c_A and c_B . The mass also has a drag force proportional to speed through the air c_d .

The dynamic state of the system is described with $z = [x, y, v_x, v_y]'$ and the static state by $z = [x, y, 0, 0]'$.

- (a) **Basic Statics.** Given parameters and an initial guess, find an equilibrium position of this system. If you have an example you like, post it and people can compare solutions.
- (b) **Basic Dynamics.** Given parameters and an initial condition, find the motion.
- (c) **All equilibria.** Given parameters, attempt to find all equilibria. How many are there? By varying parameters, what is the most and least number of equilibrium points there can be?
- (d) **Stability of equilibrium.** Given an equilibrium point, find it's stability 2 ways: a) With a dynamic simulation near the equilibrium; and b) looking at the eigenvalues of the matrix describing the linearized equations of motion. Compare.

Make appropriate plots, animations and comparisons.